CMPG 322 - Decision Support Systems (DSS) 2
SU1 - Decision Theory 3
Decision making under risk



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Outline

1. Expected monetary value (EMV)
2. Expected value of perfect information (EVPI)
3. Expected opportunity loss (EOL)
4. Sensitivity Analysis
5. Practical Exercise
6. Summarising
7. Questions

Expected monetary value (EMV)

$$EMV \times (alternative) = \sum X_i P(X_i)$$

where

- X_i = payoff for the alternative in state of nature i
- $P(X_i)$ = probability of achieving payoff (X_i) (i.e., probability of state of nature i)

EMV example (Thompson Lumber Co.)

	STATE OF NATURE	
	FAVORABLE MARKET	UNFAVORABLE MARKET
ALTERNATIVE	($)	($)
Construct a large plant	200,000	−180,000
Construct a small plant	100,000	−20,000
Do nothing	0	0

TABLE 3.1 Decision Table with Conditional Values for Thompson Lumber

EMV (alternative i) =(payoff of first state of nature) × (probability of first state of nature)
+ (payoff of second state of nature) × (probability of second state of nature)
+ ... + (payoff of last state of nature) × (probability of last state of nature)

EMV example (Thompson Lumber Co.)

- Each market outcome has a probability of occurrence of 0.50.
- Which alternative would give the highest EMV?

thus:

$$EMV\ (large\ plant) = (\$200,000)(0.5) + (-\$180,000)(0.5) = \$10,000$$

$$EMV\ (small\ plant) = (\$100,000)(0.5) + (-\$20,000)(0.5) = \$40,000$$

$$EMV\ (do\ nothing) = (\$0)(0.5) + (\$0)(0.5) = \$0$$

EMV example (Thompson Lumber Co.)

TABLE 3.9 Decision Table with Probabilities and EMVs for Thompson Lumber

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	EMV ($)
Construct a large plant	200,000	−180,000	10,000
Construct a small plant	100,000	−20,000	40,000 Best EMV
Do nothing	0	0	0
Probabilities	0.50	0.50	

Expected value of perfect information (EVPI)

- EVPI places an upper bound on what you should pay for additional information.
- EVwPI is the long run average return if we have perfect information before a decision is made.

$$EVwPI = \sum(best\ payoff\ in\ state\ of\ nature\ i)(probability\ of\ state\ of\ nature\ i)$$

Expected value of perfect information (EVPI)

EVwPI = (best payoff for first state of nature) × (probability of first state of nature)
+ (best payoff for second state of nature) × (probability of second state of nature)
+ ... + (best payoff for last state of nature) × (probability of last state of nature)

EVPI = EVwPI − Best EMV

Expected value of perfect information (EVPI)

TABLE 3.10 Decision Table with Perfect Information

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	EMV ($)
Construct a large plant	200,000	−180,000	10,000
Construct a small plant	100,000	−20,000	40,000
Do nothing			
With perfect information	200,000	0	100,000
Probabilities	0.50	0.50	

EVwPI = ($200,000)(0.50) + ($0)(0.50) = $100,000

EVwPI

Expected value of perfect information (EVPI)

TABLE 3.10 Decision Table with Perfect Information

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	EMV ($)
Construct a large plant	200,000	−180,000	10,000
Construct a small plant	100,000	−20,000	40,000
Do nothing	0	0	0
With perfect information	200,000	0	100,000
			EVwPI
Probabilities	0.50	0.50	

Expected value of perfect information (EVPI)

$$\begin{aligned}\text{EVPI} &= \text{EVwPI} - \text{Maximum EMV} \\ &= \$100,000 - \$40,000 \\ &= \$60,000\end{aligned}$$

So the maximum Thompson should pay for the additional information is \$60,000

Expected opportunity loss (EOL)

Expected opportunity loss (EOL) is the cost of not picking the best solution

- Construct an opportunity loss table;
- For each alternative, multiply the opportunity loss by the probability of that loss for each possible outcome and add these together;
- Minimum EOL will always result in the same decision as maximum EMV;
- Minimum EOL will always equal EVPI.

Expected opportunity loss (EOL)

TABLE 3.6 Determining Opportunity Losses for Thompson Lumber

STATE OF NATURE	
FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)
200,000 – 200,000	0 – (–180,000)
200,000 – 100,000	0 – (–20,000)
200,000 – 0	0 – 0

Expected opportunity loss (EOL)

TABLE 3.7 Opportunity Loss Table for Thompson Lumber

	STATE OF NATURE	
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)
Construct a large plant	0	180,000
Construct a small plant	100,000	20,000
Do nothing	200,000	0

EOL (large plant) = (0.50)($0) + (0.50)($180,000) = $90,000

EOL (small plant) = (0.50)($100,000) + (0.50)($20,000) = $60,000

EOL (do nothing) = (0.50)($200,000) + (0.50)($0) = $100,000

Expected opportunity loss (EOL)

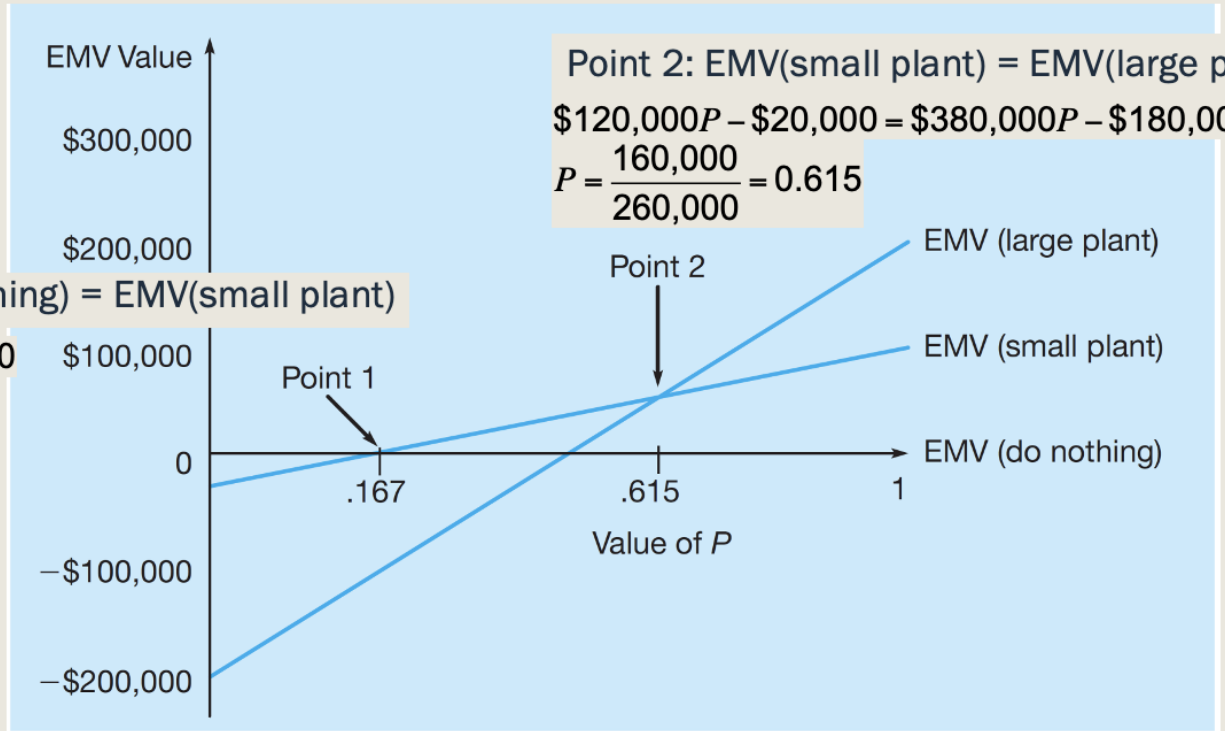
TABLE 3.11 EOL Table for Thompson Lumber

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	EOL ($)
Construct a large plant	0	180,000	90,000
Construct a small plant	100,000	20,000	60,000 ← Best EOL
Do nothing	200,000	0	100,000
Probabilities	0.50	0.50	

Define P = probability of a favorable market

- EMV(large plant) = \$200,000 P − \$180,000 (1–P)
  = \$200,000P –\$180,000 + \$180,000P
  = \$380,000P –\$180,000
- MV(small plant) = \$100,000 P − \$20,000 (1–P)
  = \$100,000P –\$20,000 + \$20,000P
  = \$120,000P –\$20,000
- MV(do nothing) = \$0P − \$0(1–P)
  = \$0

FIGURE 3.1 Sensitivity Analysis



Point 1: EMV(do nothing) = EMV(small plant)

$$0 = \$120{,}000P - \$20{,}000$$

$$P = \frac{20{,}000}{120{,}000} = 0.167$$

Point 2: EMV(small plant) = EMV(large plant)

$$\$120{,}000P - \$20{,}000 = \$380{,}000P - \$180{,}000$$

$$P = \frac{160{,}000}{260{,}000} = 0.615$$

BEST ALTERNATIVE	RANGE OF *P* VALUES
Do nothing	Less than 0.167
Construct a small plant	0.167 – 0.615
Construct a large plant	Greater than 0.615

- The Thompson Lumber Co. example is a maximisation problem.
- A minimisation problem is demonstrated in the textbook on pp 91-93.
- Solve this problem using Excel QM as an exercise.

Summarising

1. Expected monetary value (EMV)
2. Expected Value of Perfect Information (EVPI)
3. Expected Opportunity Loss (EOL)
4. Sensitivity Analysis

Questions

